

Date: 07 July 2026	Team ID: SB-7429
Project Name: Smart Lender: Applicant Credibility Prediction for Loan Approval	Maximum Marks: 4 Marks

SAMPLE PROJECT DOCUMENTATION TEMPLATE

Sample Project Documentation

1. Executive Abstract:

This project introduces Smart Lender, an end-to-end machine learning system designed to automate loan application credibility screening. By parsing 11 applicant attributes, the system leverages an optimized XGBoost classifier (83.7% accuracy) to render immediate underwriting decisions. The web interface is deployed using Flask and structured around a high-contrast dark fintech UI, providing loan risk visibility.

2. Data Imputation Summary:

Missing categorical values (Gender, Married, Dependents, Self_Employed, Credit_History) are imputed using the column mode. Numerical missing values (LoanAmount, Loan_Amount_Term) are filled using column means, ensuring data consistency.

3. Model Benchmarking Evaluation:

Model Classifier Name	Training Accuracy Status	Testing Accuracy Status
Decision Tree	82.5%	82.1%
Random Forest	80.9%	85.4%
K-Nearest Neighbors	81.7%	84.6%
XGBoost (Production Selected)	81.5%	83.7%